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Highly insulating ultrathin SiO₂ film grown by photooxidation

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Highly insulating ultrathin SiO₂ films (<5 nm) were grown by means of vacuum ultraviolet photooxidation, by use of 126, 172, and 222 nm wavelength photons generated by excimer lamps. The ultrathin silicon dioxide films were grown onto the Si(001) substrate, at low temperature (<500 °C). We found that the densities of photooxidized SiO₂ films were 13%–35% higher than that of thermally oxidized SiO₂ film. The density profile was obtained by x-ray reflectivity and showed a remarkable decrease in the SiO_x (suboxide) layer thickness at the SiO₂–Si interface, in comparison with thermally oxidized conventional oxides. The Si 2*p* photoelectron spectrum confirmed that the SiO_x layer was negligibly thin. High insulating performance of the SiO₂ film was demonstrated. © 2003 American Institute of Physics. [DOI: 10.1063/1.1597940]

I. INTRODUCTION

Reduction in length scales of gate dielectric films for deep-submicron devices is urgently needed. However, as is well known, a thermally oxidized SiO₂ film is formed with a SiO_x layer, the so-called “transition layer,” at the Si–SiO₂ interface. Poorly oxidized layers with typical thicknesses of 1–1.5 nm cause deterioration in insulating performance. Since the insulating limit is dependent on the thickness of the SiO_x layer, the thickness of SiO_x layer becomes the key factor for determining the practical thickness of SiO₂ gate dielectric film. The limiting thickness of thermally oxidized dielectric film depends on the SiO_x layer thickness.^{1–4} In order to improve the insulating of materials beyond the thickness limit of SiO₂,⁵ high-*k* (high dielectric constant) materials are intensively investigated.

Meanwhile, attempts to reduce the thickness of SiO_x layer have also made use of rapid thermal,⁶ plasma,^{7–11} ozone,^{12–18} and photo^{19–28} methods. However, SiO₂ films grown by the rapid thermal, plasma, and ozone methods suffer from surface damage and defect formation. Kazor, Boyd, and Gwillam^{12,13,14} reported low-temperature growth ultraviolet (UV)-enhanced oxidization on silicon by use of an ozonization with UV photons. Direct UV oxidation was reported to grow silicon germanium (SiGe)¹⁶ and indium phosphide (InP).¹⁷ We note that the recent work on photooxidized silicon has been aimed at achieving a high rate of deposition.^{19–28} Since very large-scale integrated (VLSI) circuit devices require an ultrathin (<5 nm) SiO₂ film, we consider that minimizing the thickness of the suboxide layer is more important than increasing the rate of oxide growth.

High-*k* gate dielectric film growth on Si involves the problem of interdiffusion during the formation of the interface SiO₂ layer, whereby the effective oxide thickness is increased. However, growing SiO₂ prior to the deposition of the high-*k* dielectric gate film has been reported to reduce the mobility of oxygen atoms.²⁹ Therefore, the growth of SiO_x

free of the ultrathin SiO₂ film is an important issue for enabling the VLSI industry to exceed the present limits.

In this article, we report our effort to reduce the suboxide layer thickness by taking a different approach, i.e., dilatory oxide growth by ambient-pressure vacuum ultraviolet (VUV) photooxidation, as well as the properties of SiO₂ films that we produced.

Thermally oxidized “conventional” SiO₂ has a density of 2.24–2.20 g/cm³; however, as is well known, SiO₂ can assume a polymorphic form having high density. In amorphous SiO₂, the Si–O bonding state has ring statistics, i.e., consists of rings of different sizes. In view that the statistics affect the packing fraction density, under certain cases where the statistics reduce the voids, the average density is expected to decrease. Although crystal polymorphs, such as cristobalite, where the oxygen atoms are located in all tetrahedral Si–Si bonds of amorphous or crystalline SiO₂ are prepared by high-pressure experiments, they are also formed by electronic excitation during growth, such as sputtering in low-temperature plasma.

II. EXPERIMENT

In photooxidized silicon dioxide growth, we used excimer lamps which can emit discrete energy photons by choosing a suitable similar rare gas or mixing with halogen; i.e., $\lambda = 126$ nm for Ar₂⁺, $\lambda = 172$ nm for Xe₂⁺, and $\lambda = 222$ nm for KrCl⁺. The silicon substrate temperature was controlled so as to maintain the growth temperature below 500 °C. The silicon wafer, a 4 in. boron doped *p*-type Si(001) wafer whose resistivity is 14–22 Ω cm, was chemically cleaned. The native oxide was removed by diluted hydrogen fluoride prior to insertion into the growth chamber. In the present study, excimer lamps with wavelengths of $\lambda = 126$, 172, and 222 nm were used. The growth chamber was designed to be high-vacuum compatible. Before oxygen gas was introduced, the growth chamber was evacuated to 0.1 Pa. The flow rate of pure O₂ gas was monitored and maintained constant during the oxidization. The oxygen partial pressure in the growth chamber was kept at atmospheric

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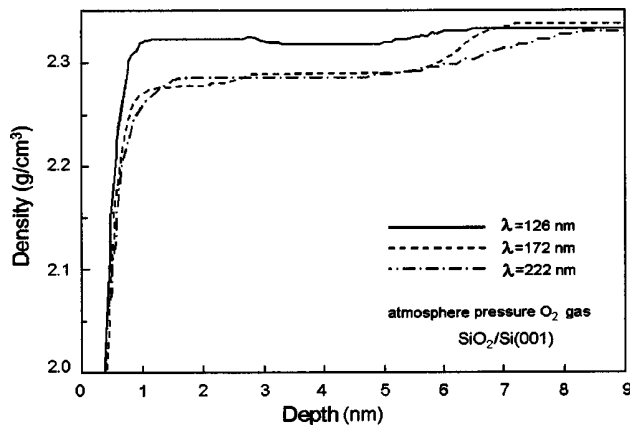


FIG. 1. Depth profile from x-ray reflectivity for $\lambda = 126$, 172, and 222 nm excimer lamps. The growth conditions of photooxidized samples are $T_{\text{sub.}} = 390^\circ\text{C}$ for 7.5 h at $\lambda = 126$ nm, $T_{\text{sub.}} = 430^\circ\text{C}$ for 5 h at $\lambda = 172$ nm, and $T_{\text{sub.}} = 390^\circ\text{C}$ for 5 h at $\lambda = 222$ nm. The densities of thermally SiO_2 thick films and bulk Si are shown in the inset.

level. The produced ultrathin (<5 nm) SiO_2 films were analyzed by x-ray photoelectron spectroscopy (XPS),^{30–32} transmission electron microscopy (TEM), x-ray reflectivity, and ellipsometry. The growth rate was estimated by x-ray reflectivity, which is used in calibration for controlling the growth. The TEM technique was used to evaluate the interface sharpness, and XPS was used to estimate the transition layer thickness for the average silicon valence.

III. RESULTS AND DISCUSSION

A. Density of UV photooxidation SiO_2 film

Figure 1 shows the depth profile of SiO_2 density obtained from x-ray reflectivity measurements for samples grown by use of photons of various wavelengths. Grazing incidence x-ray reflectometry (GIXR), measured over wide ranges in angle and intensity, contains information on density and roughness. Assuming the SiO_2 layer has a graded density along the direction normal to the surface, the parametrized density distribution can be obtained by a square fit to the x-ray reflectivity based on the Parratt method.³³ Furthermore, Hashimoto *et al.*^{34,35} determined the suitable parameter for an ultrathin SiO_2 layer. Consequently, we used their developed technique and determined the depth profile of the photooxidized SiO_2 density.

The growth conditions of SiO_2 film for depth profile of SiO_2 density were $\lambda = 126$ nm photooxidation for 7.5 h at a substrate temperature ($T_{\text{sub.}}$) of 390°C , $\lambda = 172$ nm for 5.0 h at $T_{\text{sub.}} = 430^\circ\text{C}$, and $\lambda = 222$ nm for 5.0 h at $T_{\text{sub.}} = 390^\circ\text{C}$. The density of conventional thermally oxidized SiO_2 is 2.20 g/cm^2 for wet oxidation and 2.24 g/cm^2 for dry oxidation. The average density (D_{av}) of $\lambda = 172$ and 222 nm photooxidized SiO_2 film is 2.27 g/cm^2 and that for $\lambda = 126$ nm is $D_{\text{av}} = 2.32\text{ g/cm}^2$. Figure 1 shows that all photooxidized SiO_2 film exhibits D_{av} higher than that of conventionally thermally oxidized SiO_2 film. The $\lambda = 126$ nm photooxidized film has an especially high value.

The transition is observed as a slope change at the interface for $\lambda = 172$ and 222 nm photooxidized SiO_2 film, but for

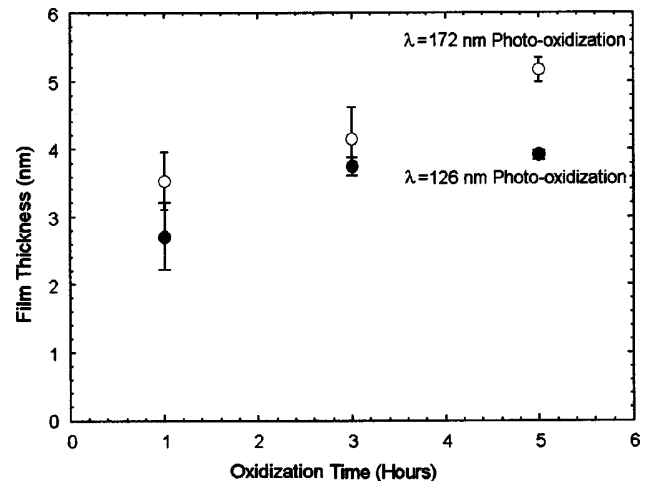


FIG. 2. Thickness of silicon dioxide film grown on a silicon substrate by photooxidation at $\lambda = 126$ and 172 nm, as a function of irradiation time.

$\lambda = 126$ nm a slope change is hardly observed, because of the dense form of SiO_2 in comparison with silicon formed by photooxidation (2.33 g/cm^2).

The density profile exhibits a remarkable wavelength dependence between $\lambda = 126$ and 172 nm. The samples oxidized at $\lambda = 172$ and 222 nm exhibit a small difference in D_{av} , but differ considerably in slope change in the transition layer. The density variation of the transition layer for $\lambda = 222$ nm photooxidized SiO_2 is smaller in magnitude than that of $\lambda = 172$ nm.

In summary, GIXR measurements show that VUV photons of higher energy formed oxide films exhibiting higher density and sharper interface. Therefore, the mechanism of densification and the interface formation are both related to the excited state of oxygen atoms formed by VUV photonirradiation. The SiO_2 can assume a polymorphic crystal structure of high density, such as cristobalite, which has the same density as silicon. In the case of amorphous SiO_2 , the Si–O bonding structure has ring statistics, i.e., consists of rings with different numbers of atoms. Packing fraction density increases when the statistics broaden, thereby reducing the macroscopic voids.

Figure 2 shows the thicknesses of SiO_2 films photooxidized at $\lambda = 126$ and 172 nm versus irradiation time, specifically, 1, 3, and 5 h. The substrate temperature is 460°C ($\lambda = 126$ nm) or 470°C ($\lambda = 172$ nm). The thickness of SiO_2 films appears to saturate with increasing irradiation time. Also, the lower the photon energy, the higher the growth rate, although this could result from substrate temperature dependence. In order to evaluate the kinetics in detail, the photon flux dependence is currently being investigated.

B. Surface and interface of photooxidized SiO_2 film

Figure 3 shows a TEM cross section image of a sample SiO_2 film that had been photooxidized at $\lambda = 126$ nm on a Si(001) substrate. The SiO_2 film covered aluminum that had been grown by vacuum evaporation, which was used as an electrode for $J-V$ (current density–voltage characteristic)

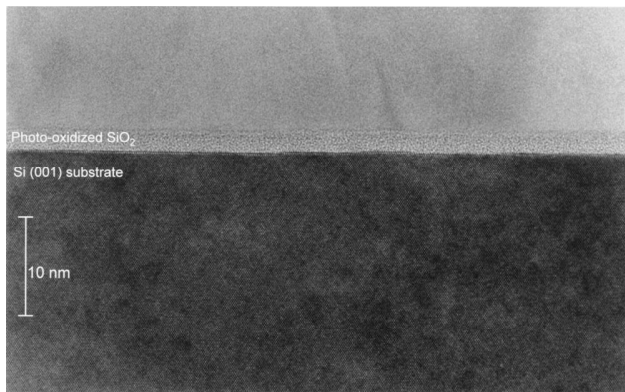


FIG. 3. TEM cross-section image of photooxidized sample. The growth conditions of SiO_2 film were $\lambda = 126$ nm photooxidation for 5 h at a substrate temperature of 390 °C.

measurements. The sample was prepared by the ion-milling method. The TEM image was obtained by H-9000NAR (Hitachi) having an electron acceleration voltage of 200 kV. For high-resolution measurement, the electron beam along $\langle 110 \rangle$ was used. The film thickness is 3.5–3.7 nm, which is obtained after calibration using the lattice spacing of silicon along $\langle 111 \rangle$. The interface roughness is the same as the initial surface roughness of silicon; from this we concluded that the growth of SiO_2 does not degrade the interface roughness.

C. Oxygen stoichiometry in photooxidized SiO_2 film

In SiO_2 , where silicon atoms are tetrahedral bonded with oxygen atoms, the silicon valence is Si^{4+} , whereas in suboxides, Si assumes lower valence states, such as Si^{3+} , Si^{2+} , and Si^{1+} , depending on the lack of oxygen coordination. In constant, bulk silicon takes the neutral Si^0 state. Utilizing the chemical shift of photoelectron spectroscopy, one can evaluate the average chemical composition within the probe region. The XPS data were obtained at 45° takeoff angle. The x-ray source is monochromerized $\text{AlK}\alpha$ line (1486.6 eV), and the energy resolution is 0.52 eV. The beam diameter is 100 μm ϕ , and detection depth is about 5 nm at 45° takeoff angle. The background was determined by taking into consideration the energy loss from inelastic scattering. The employed XPS system was PHI Quantum 2000.

Figure 4 shows the $\text{Si}2p_{1/2,3/2}$ signal profile. Between the bulk Si (Si^0) and silicon dioxide (Si^{4+}) signals, a small amount of suboxide contribution (Si^{1+} and Si^{2+}) is observed. The signals were fitted by a Gaussian function and the fraction was estimated to be less than 5%, which is comparable with maximum error. The results of curve fitting are summarized in Table I. The results show that the suggestion of suboxide thickness in the photooxidized SiO_2 film from the XPS result is remarkably small.

The SiO_2 thickness (d_{SiO_2}) can be calculated from Si 2p signal intensity of XPS data.^{36,37}

$$d_{\text{SiO}_2} = l_{\text{SiO}_2} \sin \alpha \ln \left[\frac{I_{\text{Si}^{4+}}/I_{\text{Si}^0}}{I_0/I_\infty} + 1 \right],$$

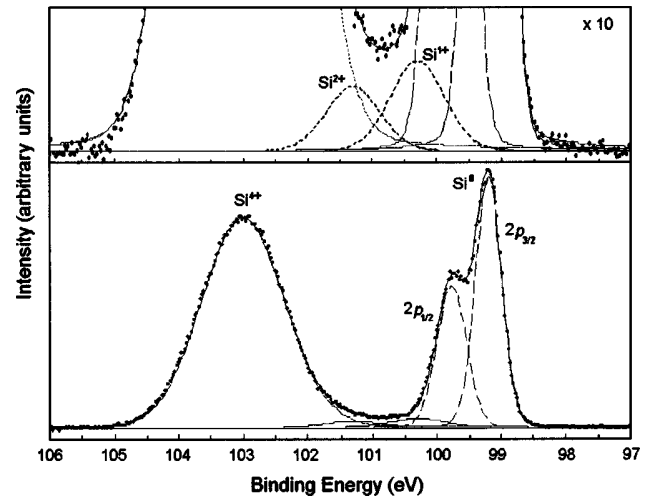


FIG. 4. XPS data of 2p spectrum of $\lambda = 126$ nm photooxidation. The open circles are as-recorded data after background subtraction. The dashed lines are fitting curves for each combined state by a Gaussian function. The upper graph is scaled up tenfold from the base graph.

where l_{SiO_2} is the photoelectron mean-free path in the SiO_2 film and α is the photoelectron takeoff angle. The conventional SiO_2 film of l_{SiO_2} is determined as 2.6×10^{-7} cm for $\text{AlK}\alpha$.³⁸ $I_{\text{Si}^{4+}}$ and I_{Si^0} are integrated photoelectron intensity of SiO_2 and bulk silicon. The experimental intensity ratio I_∞/I_0 is determined as 0.82 for the $\text{AlK}\alpha$ line. The number of silicon atoms in suboxidation states is taken as

$$N_{\text{SiO}_x} \approx n_{\text{Si}^0} l_{\text{Si}^0} \frac{I_{\text{Si}^{x+}} \sigma_{\text{Si}^0}}{I_{\text{Si}^0} \sigma_{\text{Si}^{x+}}} \sin \alpha,$$

where n_{Si^0} is the atomic density of silicon, which is $5.0 \times 10^{22} \text{ cm}^{-3}$ and l_{Si^0} is the photoelectron mean-free path in Si, which is determined as 1.6×10^{-7} cm for $\text{AlK}\alpha$.³⁸ $\sigma_{\text{Si}^{x+}}$ and σ_{Si^0} are the photoionization cross section and are assumed as the SiO_x and Si states at high photon energy in XPS. The results of calculation from our data are given as $d_{\text{SiO}_2} = 3.0$ nm and $\sum_{x=1}^3 N_{\text{Si}^{x+}} = 4.0 \times 10^{14} \text{ atoms/cm}^2$.

D. Electrical characteristics

Figure 5 shows the J – V characteristic curves of $\lambda = 126$ and 172 nm photooxidized films. The samples were formed into a metal-insulator-semiconductor structure and measured by the two-terminal method. Photooxidation at $\lambda = 126$ and 172 nm produced SiO_2 films having thicknesses of $d_{\text{SiO}_2} = 3.5$ and 3.0 nm and breakdown voltages of about

TABLE I. Intensity, chemical shifts, and (full width at half maximum value (FWHM) of each chemical state of Si 2p signal.

Chemical state	Chemical shift (eV)	FWHM (eV)	Fraction (%)
Si^0 (Si)	0	0.49	23.20
Si^{1+} (Si_2O)	1.1	1.00	1.70
Si^{2+} (SiO)	2.1	1.00	1.21
Si^{3+} (Si_2O_3)	3.0	1.00	0
Si^{4+} (SiO_2)	3.8	1.51	59.47

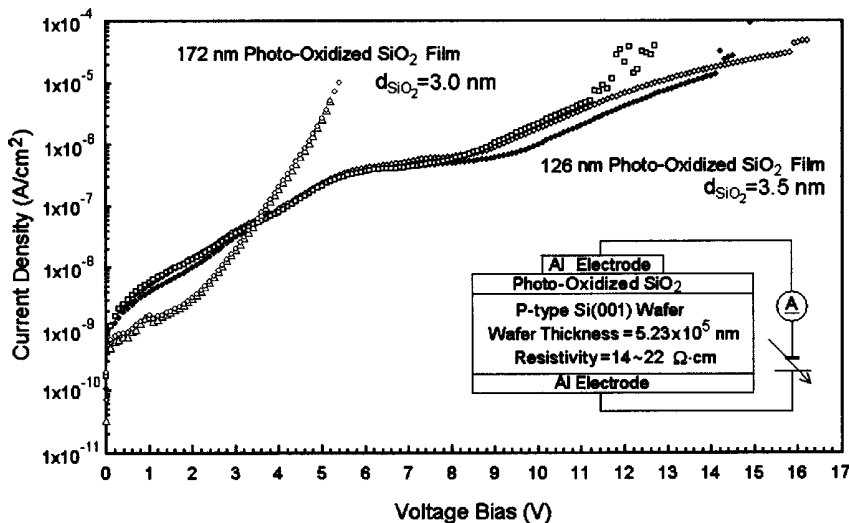


FIG. 5. J - V characteristics for $\lambda = 126$ and 172 nm photooxidized SiO_2 film grown on the p -type $\text{Si}(001)$ substrate. The substrate conditions and measuring method are as specified in the inset.

15.5 and 5.5 V. Note that the breakdown voltage is higher for $\lambda = 126$ nm photooxidation film. Estimated electrical field strength of $\lambda = 126$ and 172 nm photooxidized SiO_2 film is about 45 and 18 MV/cm. The leakage current of $\lambda = 126$ and 172 nm photooxidation film is 10^{-4} – 10^{-10} A/cm² from zero to breakdown voltage. The results clearly demonstrate the superior insulating performance of photooxidized ultrathin SiO_2 film, as compared with that reported for conventional oxide ultrathin SiO_2 films.³⁹ The insulating performance of photooxidized SiO_2 film cannot be explained by simply increasing density by 10%–30%. Rather, we believe that the thickness of the SiO_x layer or the number of oxygen defects is of prime importance. The nature of the SiO_2 –Si interface and the relation with insulating performance is currently under investigation and will be reported elsewhere.

IV. CONCLUSION

We have grown highly dielectrical ultrathin SiO_2 film on $\text{Si}(001)$ substrate by use of a VUV photooxidized technique employing ambient pressure, and obtained findings consistent with those of earlier studies. Average thickness and interface with transition layer thickness were found to depend on wavelength. The higher the photon energy, the denser the oxide films, and for $\lambda = 126$ nm photons, the density coincides with that of silicon itself, making a continuous density profile across the SiO_2 –Si interface. Second, the interface layer thickness decreases with increasing photon energy. The average density suggests that the ring statistics in Si–O network are modified by UV photons. A considerable difference in electrical insulating performance is observed between $\lambda = 126$ and 172 nm photooxidized SiO_2 film. The breakdown voltage of $\lambda = 126$ nm photooxidized SiO_2 film is considerably higher than that of 172 nm photooxidized film, but leakage current is almost the same. We conjecture that dielectric performance of SiO_2 film relates to the ring statistics of Si–O network and the nature of the SiO_2 –Si interface. Since the Si–O bonding condition, which becomes densified, is attained by use of high photon energy, the high excitation oxygen is considered to have a beneficial effect on forming

densified SiO_2 film. The potential energy in each excitation state of an oxygen atom has been calculated.⁴⁰ However, the effective excitation oxygen state and mechanism working on oxidation remain unclear. The important feature of this article that we demonstrated is that photooxidized film exhibits electrical performance higher than that of thermally oxidized SiO_2 film.

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